Emergency Shelter & Housing Outcomes - Factors for success

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DATA 510 Data Science Capstone · Summer 2026
In collaboration with Church at the Park - https://www.church-at-the-park.org/about

Abstract

Church at the Park's (C@P) program data, including shelter and housing applications, demographics, assessments, and case notes live in 3 disconnected systems. In this project, we explored how the data can be joined together to provide insights on program delivery, average times post-application and in stays, and what is predictive of positive exits.

Through this work we hope to help Church at the Park's staff better understand their program outcomes and advocate for increased resources that will lead to more community benefit.

Introduction & Motivation

Problem

C@P provides vital services to the local community, but faces challenges in efficiently managing and analyzing the data related to these services. This limits their ability to make data-driven decisions, improve service delivery, and make a strong case for additional monetary support. This lead to the following research questions:

- Can we construct a sustainable data engineering pipeline that integrates C@P's 3 program datasets?
- What insights can we derive from integrated data to improve service delivery and support decision-making? Specifically:
  - Are there factors that are correlated to the average length of wait to get into shelter, and the average stay in shelter?
  - Are there factors that are correlated to whether a participant is successful getting into shelter, more likely to return to C@P for services, or contribute to successful exit from shelter?

Impact & stakeholders: C@P staff and leadership can use these insights to improve programs, allocate resources, and advocate for investments. Participants of the programs will benefit if the program designs are improved, or if additional resources are added.

Background & Related Work

C@P utilizes federal and state mandated databases for tracking stays, demographics, assessments, and case notes. Many of the fields are defined by the U.S. Department of Housing and Urban Development, and we relied on the organization's definitions for program exit types and their program guidelines.

Methods & Evaluation

Duration (survival) analysis: a statistical approach to evaluate how the time it takes to enter or exit shelter, and explore when this timeframe varies within stratifications of a variable.

XGBoost: a machine learning algorithm, used to explore factors that lead to successful entry to shelter, or positive exit from shelter.

Validation/evaluation methods:

- P-values for significant differences, parametric curve overlay of Weibull and Kaplan-Meier curves, shape and scale parameters.
- Log Loss; ROC-AUC Score; Confusion Matrix with F1-Score, Precision/Recall

Data, Engineering & Ethics

Data: We used 2024-2025 data from C@P, that came from three sources:

- C@P's application platform
- Case notes
- A HUD-mandated tracking system

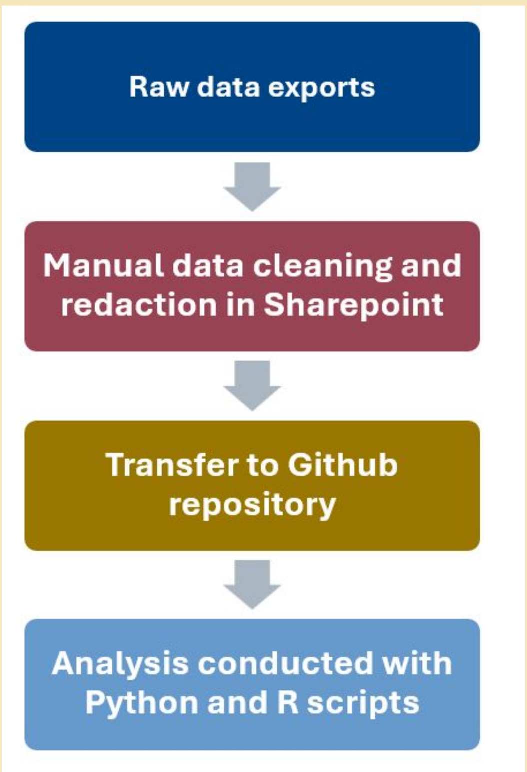
These sources were combined to create two datasets for analysis: an emergency shelter stay dataset (1,108 rows) and an application data set (5,393 rows).

Ethics & responsible use:

We protected personal information of participants by conducting identity resolution and case note language processing on the organization's server.

We also obscured sensitive features to protect against participant identification, and practiced ethical storytelling in line with the organization's standards.

Data Pipeline



Discussion, Limitations & Conclusions

Discussion: We found a mix of factors seem to affect the metrics noted in our research questions, but also found there are real world circumstances that likely affect the results and interpretation.

Limitations & future work:

- **Nature of the data & analysis:** There may be patterns in the data that likely reflect the ground realities that C@P faces. For example, time of year and availability of shelter may affect the metrics we were examining. Future work may consider implementing measures to account for these patterns.
- **Scope & causation:** We did not examine what factors lead to specific outcomes. For example, this analysis doesn't cover what factors specifically lead to obtaining rental housing versus living with a friend. Our analysis is not causative, and more work would be needed to establish any causation.
- **Generalizability:** This study only uses data for one organization over 2 years, in one specific part of the state. Our findings may not be highly generalizable to other populations.
- **Data quality:** Connecting a participant across data sources was difficult, and we likely did not match every participant across all three sources. Procedural and data entry improvements to identify a participant across all stages of C@P's processes could improve the quantity and quality of the data collected. Additionally, substantial missingness in the data may also affect our analyses.

Conclusions: Our integrated data helped identify factors that may influence a participant's experience, but they do not convey the full story of what affects the metrics we examined, and should be considered side-by-side with staff and participant experiences. Future research on outcomes, or repeating this research with other populations could provide more general and deeper insights.

Reproducibility & References

Project Website: https://stuckerwu.github.io/

Key dependencies: [Python: pandas, numpy, seaborn, scikit-learn, duckdb, matplotlib, pathlib, xgboost, shap, re] [R: tidyverse, janitor, lubridate, survival, survminer, rlang, broom, stringr, ggally]

Results & Visualizations

Factors associated with time to get into shelter, or length of stay in shelter

Fig. 1: Exits from shelter to positive housing outcome, by location

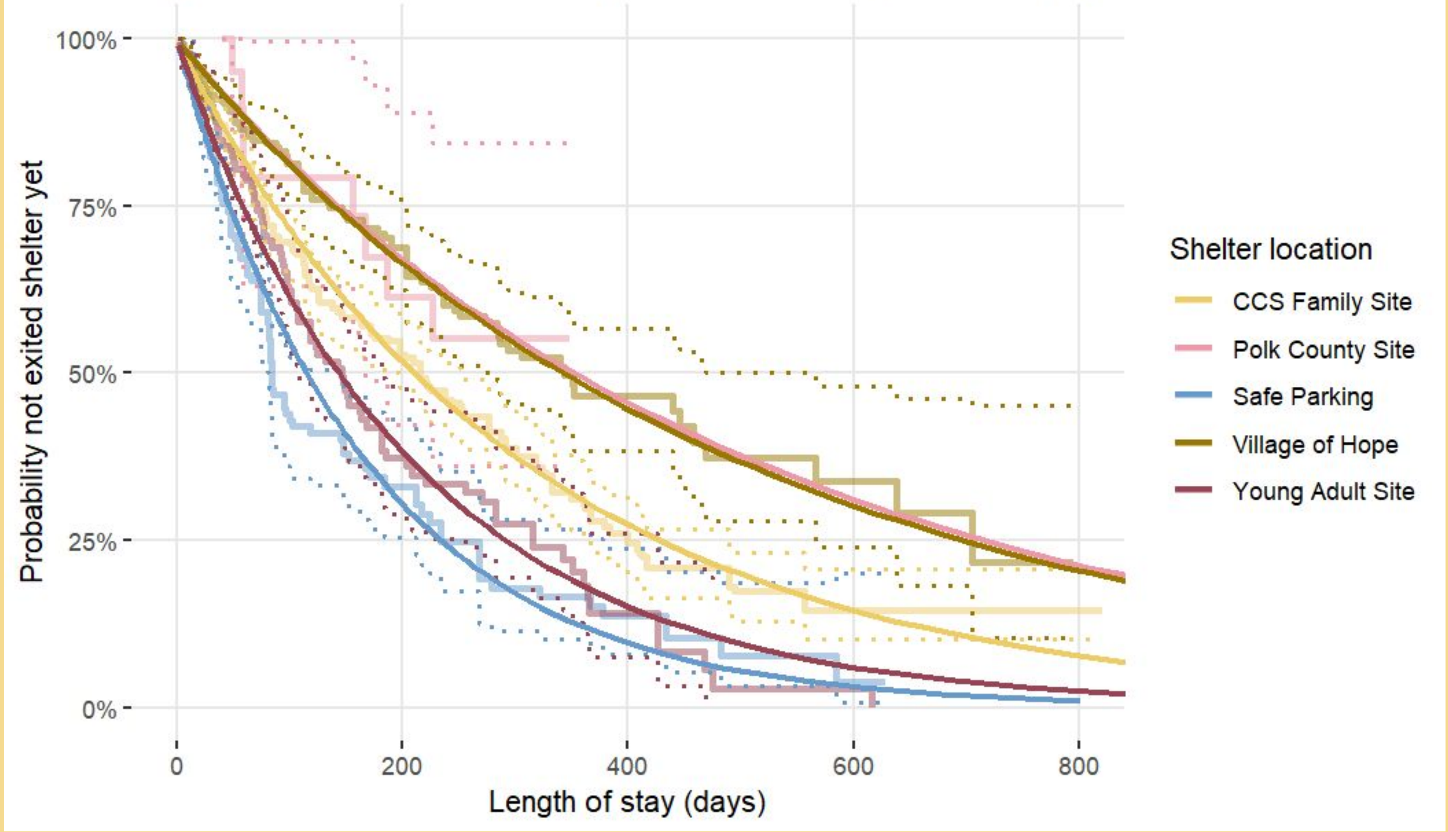
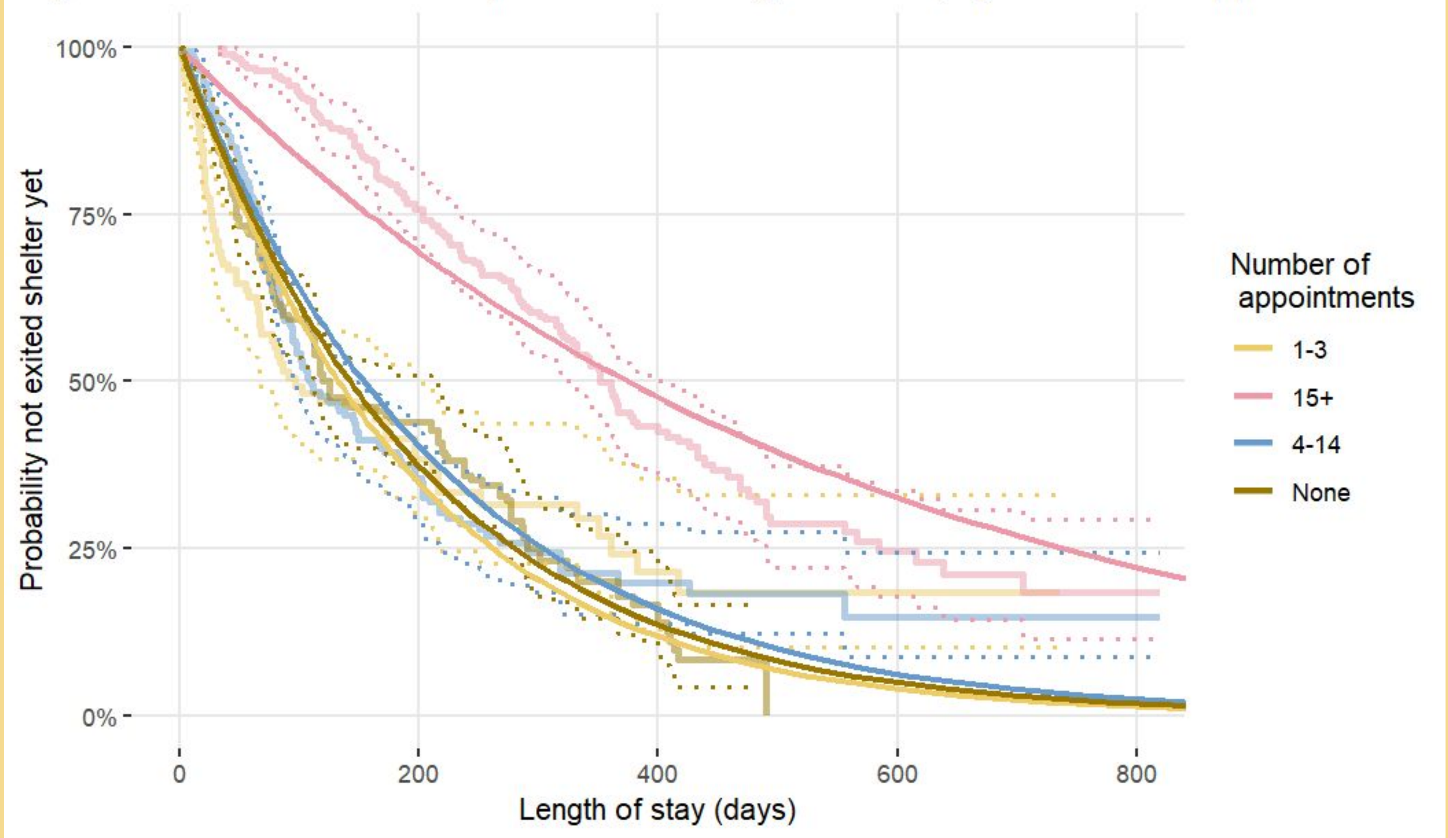
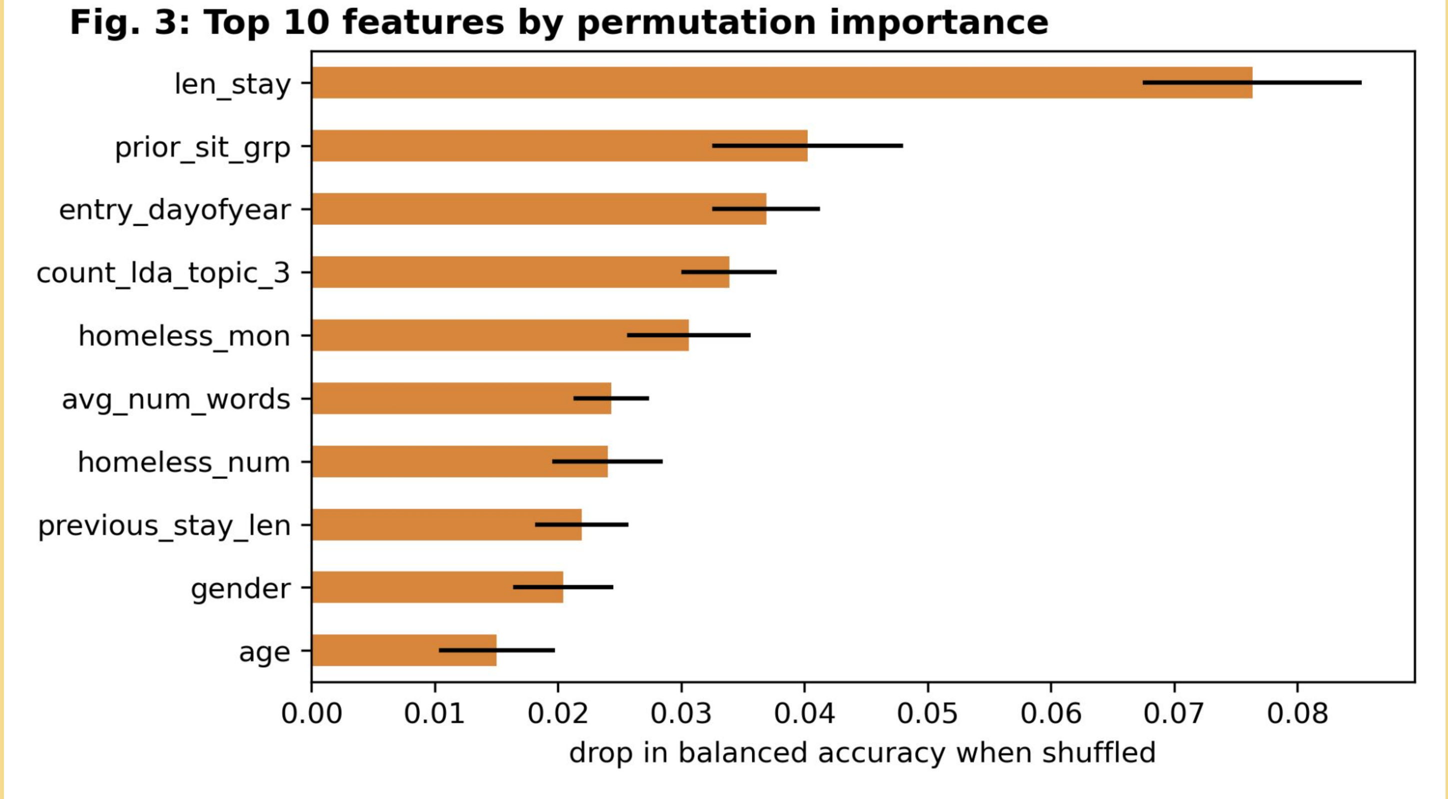


Fig. 2: Exits from shelter to positive housing outcome, by number of appointments



Factors associated with positive exits from shelter programs

Fig. 3: Top 10 features by permutation importance



Multiple factors may affect the timeframes to enter or exit shelter.

Specific circumstances or traits of the participant

We found notable differences in time spent in or waiting for shelter depending on:

- Shelter location
- Whether the participant has health coverage *(stays only)*
- Gender
- Age

Fig. 1 shows how length of stay before exiting to a positive outcome tends to be longer for the Polk County and Village of Hope sites compared to others.

Level of engagement in case management

Our data sources include case management notes that summarize assistance participants were receiving from a case manager. We found multiple variables where more engagement appears associated with longer stays in shelter. Those factors included:

- Mentions of various key words: "Appointment", "Medication", "Identification" and "Benefits."
- Length of appointments

We found that these interactions could ultimately be generalized to an "Engagement" metric, which we identified as the total number of appointments. *See Figure 2.*

Factors in Positive Exits

- Longer stays are positively associated with a positive exit, while shorter stays range from neutral to strongly negative.
- Participants coming from a 'place not meant for habitation' were more likely to have a negative exit.
- Entry in the 2nd half of the year 2024/2025) is predictive of a positive exit.
- LDA Topic 3 relates to noncompliance with site rules, and is negatively associated with a positive exit.

Factors associated with shelter entry

We built both Duration Analyses and an XGBoost model to identify factors associated with shelter entry status (entry into shelter, return application, or did not enter shelter). However, these models were not predictively powerful. We recommend future efforts attempt separate models for each program's applications.